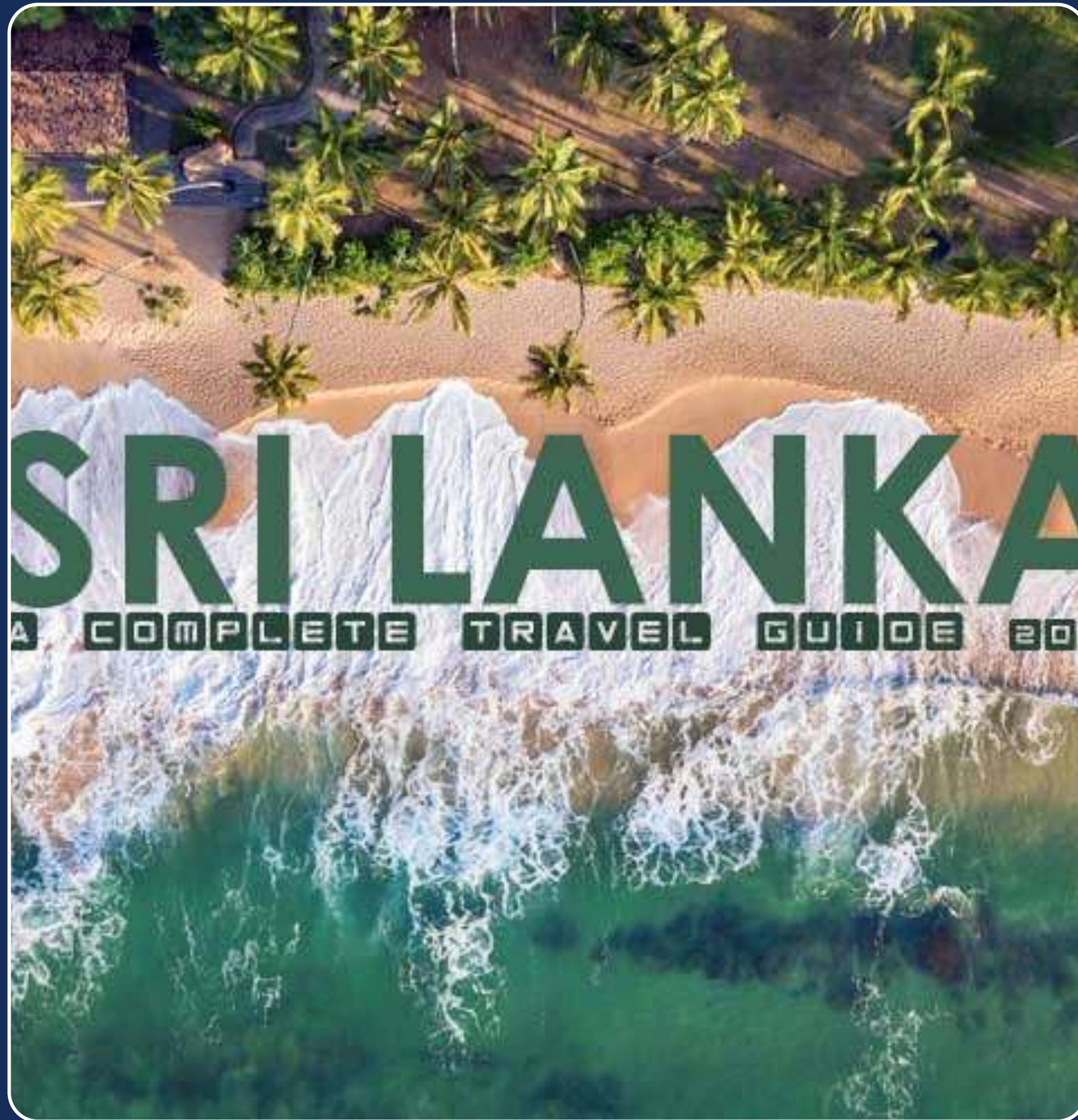




WEB-BASED TOUR GUIDE SYSTEM

AI-Powered Tourism Platform for Sri
Lanka

Project Proposal

PROJECT ID: Y2.S2.WD.DS-02-G04

Our Team

Members

- IT24400066- Athauda P.A.K.D
- IT24101887- Faizam M.F.M
- IT24102131- De Silva P.K.N
- IT24104301- Karannagoda K.V.R
- IT24101446 - Wijekoon W.M.N.S.B
- IT24101896 - Abeywickrama J.B



PROJECT SCOPE & OBJECTIVES

🌐 Project Vision

- An AI-powered web platform to help tourists plan smart personalized trips in Sri Lanka.

📖 Why This Project?

Tourists face:

- Too many travel websites & reviews
- Time-consuming manual planning
- No personalized travel suggestions



🚀 Our Smart Solution

🔑 User Accounts - Secure login & profiles

🔍 Smart Search - AI finds places by meaning

Best Routes - AI chooses the optimal route

🎁 Custom Tours - Build packages with guides or without

★ Review Summary - AI reads reviews for users

💳 Secure Payments - Safe bookings & payments



🎯 Project Objectives

- Use AI & ML for smart place search
- Apply Genetic Algorithm for route optimization
- Build AI chatbot
- Summarize reviews using NLP
- Deliver a personalized experience
- Complete project within 9 weeks



Target Users

- Tourists (Local & Foreign)
- Tour Guides
- System Admins



FUNCTIONAL REQUIREMENTS

Functional Requirements

1. User Authentication
2. Tourist Spot Browsing
3. Tour Package + Guide Selection
4. Booking Management
5. Feedback & Review System
6. Payment Management
7. AI Chat Bot

1: USER AUTHENTICATION | IT24101887

Create: User registration (Tourist / Guide / Admin)
Store encrypted user credentials and preferences

Read: User login verification

Edit: Profile details (name, interests, budget)
Update preferences based on user activity

Delete: Deactivate or remove user accounts



This component securely manages user access, supports a QA chatbot, and uses user preferences to provide personalized travel recommendations.

FEASIBILITY (2 Weeks)

Time Feasibility:

- Week 1: User authentication
- Week 2: Preference tracking & basic ML similarity

Technical Feasibility:

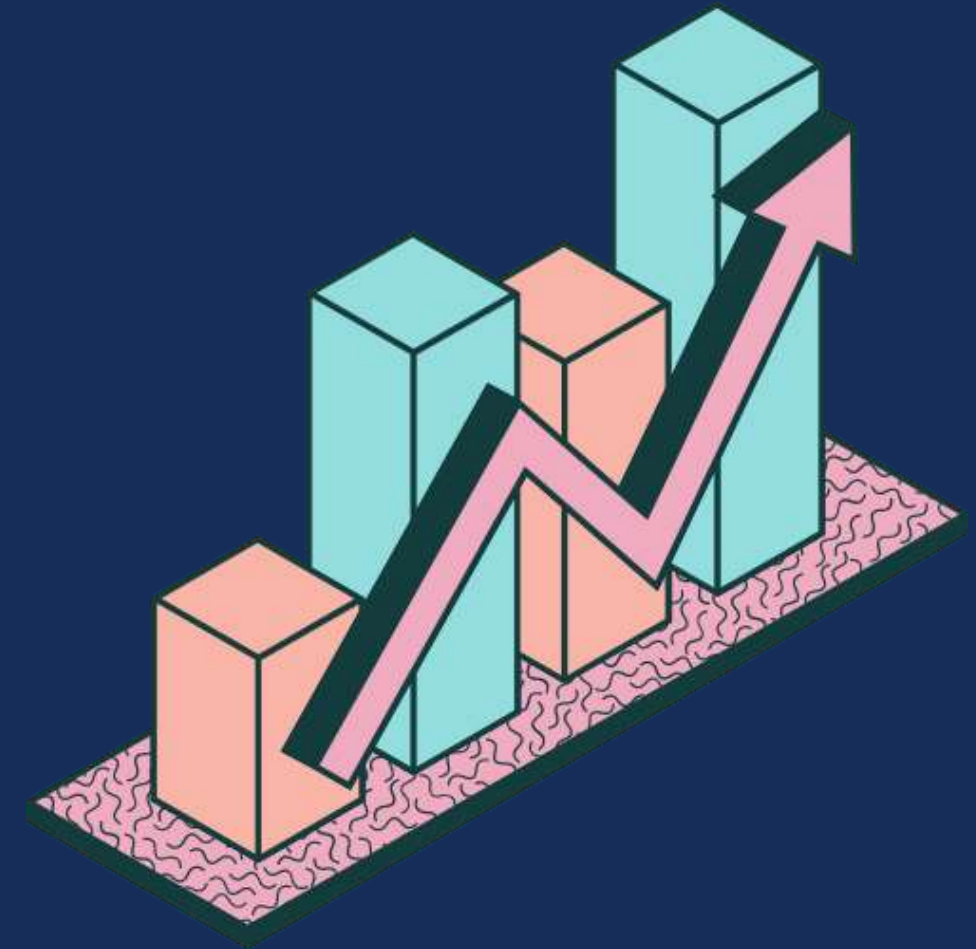
- No complex or experimental algorithms

Resource Feasibility:

- Requires only a laptop and open-source libraries

Skill Feasibility:

- Prior experience with authentication systems
- Basic ML concepts are manageable within timeline



2: Tourist Spot Browsing | IT24101896

Create: Add tourist spots with name, description, category

Read: View all spots (normal list)

Update: Modify spot details

Delete: Remove outdated



AI Functionality

Converts spot descriptions into semantic embeddings.

Matches searches with similar places, not just words

Shows the most relevant tourist spots



Feasibility Analysis (2.5 Weeks)

Time:

- Week 1: CRUD & data collection
- Week 2: Generate embeddings using pre-trained model
- Week 3: Build search API & link frontend

Technical:

- Uses pre-trained AI model (Sentence Transformer)
- Handle CRUD in manually

Resources:

- Only a laptop and free libraries needed

Skills:

- Basic Python , Java and API knowledge
- ML knowledge



3: TOUR PACKAGE & GUIDE SELECTION | IT24102131

Update: Edit package details, spots, guides

Delete: Cancelled packages

AI Functionality

optimize routes and package itineraries

Suggests guides matching package type and user preference

Provides personalized, multi-turn conversation experience



■ Feasibility Analysis (2.5 Weeks)

Time Feasibility:

- Week 1: Package & guide CRUD management
- Week 2: Chatbot implementation with API
- Week 3: Integration with search, GA, and review summaries

Technical Feasibility:

- Uses pre-built API for chatbot

Resource Feasibility:

- Only a laptop, internet, and API required

Skill Feasibility:

- Familiarity with Python & API calls sufficient



4: BOOKING MANAGEMENT | IT24400066

Update: Edit booking dates, reschedule, or change guide

Delete: Cancel bookings and remove records if allowed

AI / Optimization Functionality

GA calculates the best route based on distance, cost, duration and preferences

Tests multiple route options and picks the highest scoring one

Returns optimized itinerary with daily plan, total cost, and travel distance



Feasibility Analysis (2 Weeks)

Time Feasibility:

- Week 1: Develop & test GA algorithm
- Week 2: Implement booking CRUD system

Technical Feasibility:

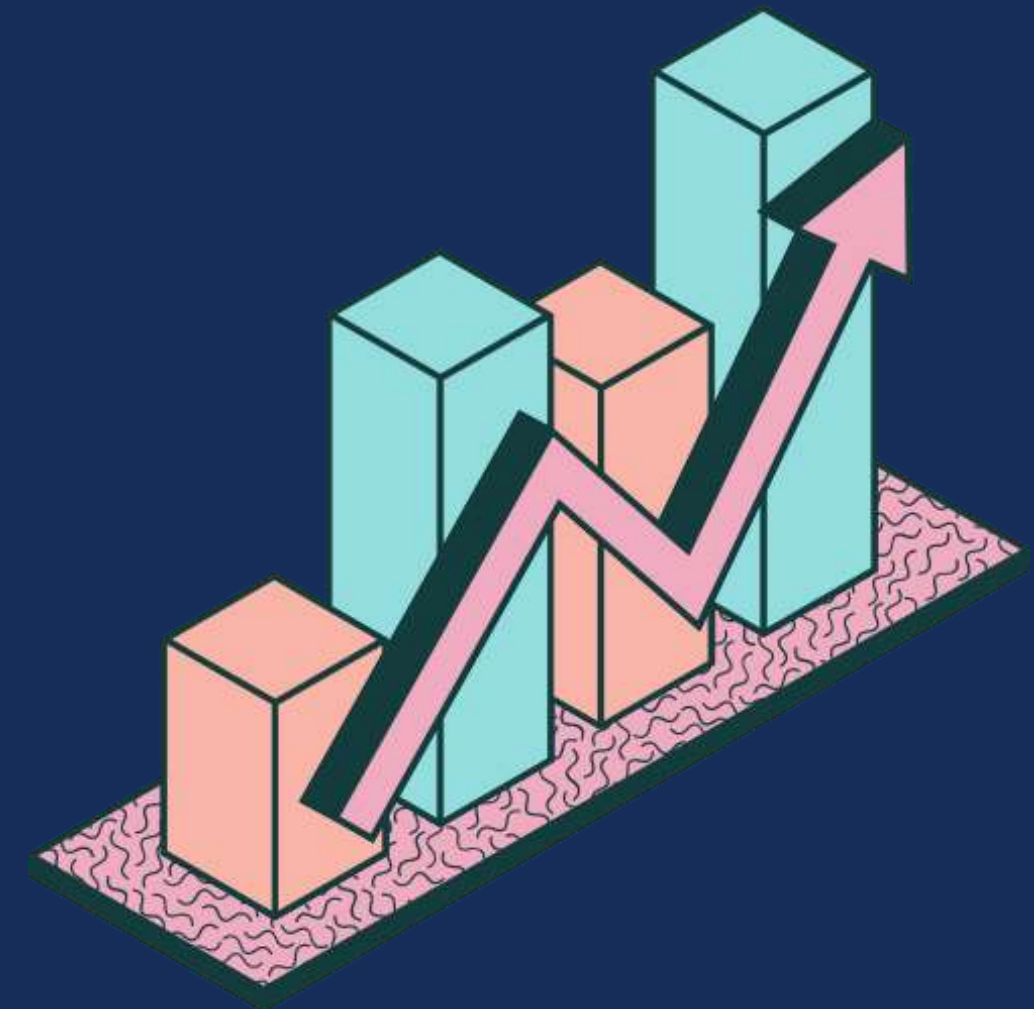
- GA implemented in Python with NumPy
- Booking API built using Flask
- PostgreSQL for storage

Resource Feasibility:

- Only laptop and open-source libraries required

Skill Feasibility:

- Python and API knowledge sufficient



Payment Management | IT24101446

- 1.Create: Add new payment record when user confirms booking
- 1.Read: View payment status and history linked to bookings
- 1.Update: Change payment status (e.g., Pending → Confirmed)
- 1.Delete: Remove test or canceled payment records



The system simulates payment processing by creating, viewing, updating, and managing payment records linked to bookings, without using a real payment gateway.

FEASIBILITY (2 Weeks)

Technical Feasibility

- Simple CRUD for payments
- No real payment gateway needed

Time Feasibility

- Week 1: Payment tracking
- Week 2: UI development

Resource Feasibility

- Free tools: React, Node.js

Skill Feasibility

- React/frontend experience enough
- Standard web dev tasks only



FEEDBACK & REVIEW SYSTEM | IT24104301

Create: Users add new reviews for spots

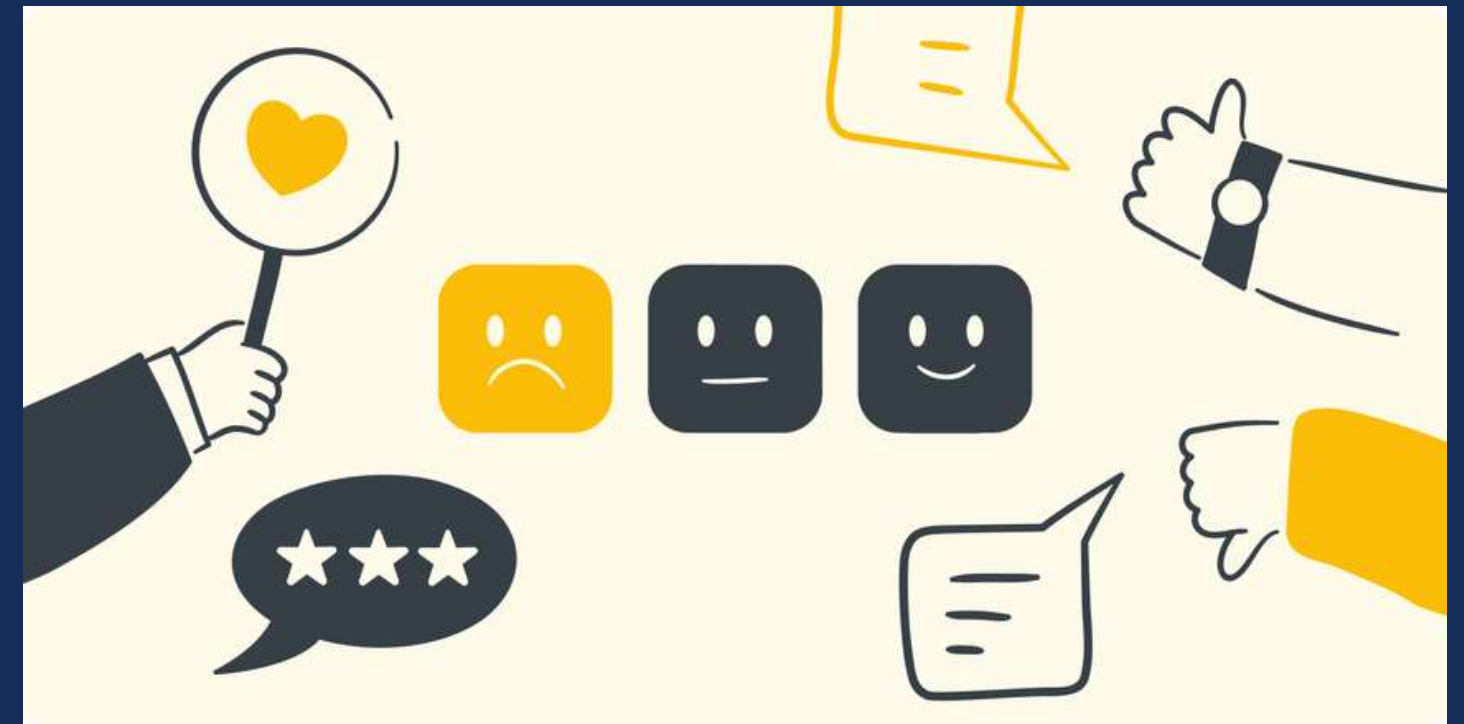
Read: View all reviews and AI-generated summaries

Update: Edit reviews if needed

Delete: Remove inappropriate or outdated reviews

AI Functionality

- Reads all reviews and finds common themes
- Analyzes sentiment
- Generates a short, smart summary with highlights, warnings, and tips



■ Feasibility Analysis (2 Weeks)

Time Feasibility:

- Week 1: Build review system & basic summarization
- Week 2: Add sentiment analysis and categorize summaries

Technical Feasibility:

- Uses pre-trained NLP model

Resource Feasibility:

- Only a laptop and free (HuggingFace) library required
- No paid services or extra hardware

Skill Feasibility:

- Basic knowledge of Python and NLP enough



NON-FUNCTIONAL REQUIREMENTS

1. PERFORMANCE REQUIREMENTS

2. SECURITY REQUIREMENTS

3. RELIABILITY & AVAILABILITY

4. USABILITY REQUIREMENTS

**6. MAINTAINABILITY
REQUIREMENTS**

9-WEEK PROJECT TIMELINE

- 
- 1** — **Weeks 1-3: Foundation**
Architecture design, database setup, authentication system, and initial ML embeddings for smart search functionality
 - 2** — **Weeks 4-5: Intelligence Layer**
Complete search API, collaborative filtering, genetic algorithm optimization, and sentiment analysis integration
 - 3** — **Weeks 6-7: Integration**
Connect all components, conduct comprehensive API testing, and optimize performance across the full system
 - 4** — **Weeks 8-9: Launch Ready**
Final testing, cloud deployment, documentation completion, and demo preparation for project presentation

TEAM COLLABORATION

Meetings:

- Weekly team meeting (Monday) – progress & planning
- Daily short standups during integration phase

Tools:

- GitHub – code sharing & version control
- Teams / Slack – quick communication

Documentation

- Google Drive – shared files
- Meeting minutes & README files



Meeting Minutes

Minutes

Board meeting

DATE

January 27th

TIME

9:30AM

MEETING CALLED TO ORDER BY

Lecturer

IN ATTENDANCE

IT Number	Name
IT24400066	Athula P.A.K. D
IT24104301	Karunagoda K.V. R
IT24101887	Fazam M.F.M
IT24101446	Wijekoon W.M.N.S. B
IT24102131	De Silva P.K. N
IT24101896	Abeywickrama J. B

APPROVAL OF MINUTES

This was our first project meeting, so there were no previous minutes to approve.

BOARD

The lecturer asked everyone to sit according to their groups. After forming the groups, members introduced themselves and got to know each other.

REPORTS

The group discussed possible project topics from different perspectives. We also searched for relevant datasets by visiting Kaggle and Hugging Face. The focus of this meeting was selecting a suitable project topic and finding related datasets.

ANNOUNCEMENTS

It was decided to conduct another meeting to finalize the selected project and discuss further details.

NEXT MEETING

January 27th, Lecture Hall 0608

Minutes

Board meeting

DATE

January 27th

TIME

12:30 PM

MEETING CALLED TO ORDER BY

Team Members

IN ATTENDANCE

IT Number	Name
IT24400066	Athula P.A.K. D
IT24104301	Karunagoda K.V. R
IT24101887	Fazam M.F.M
IT24101446	Wijekoon W.M.N.S. B
IT24102131	De Silva P.K. N
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APPROVAL OF MINUTES

Minutes from the earlier meeting on the same day at 9:30 AM were noted and accepted.

BOARD

The lecturer guided the group to continue working on project topics and datasets.

REPORTS

- The group discussed the previously shortlisted topics and selected suitable datasets for each.
- For one topic, **"Predicting Energy Usage for a Smarter Future"**, the group prepared an 8-slide presentation.
- The presentation was delivered to the class, and feedback was received from the lecturer.
- Based on the feedback, it was decided that a new topic should be considered since the project mainly focuses on AIML components.
- Additional topics were discussed, and relevant datasets were identified for them.

ANNOUNCEMENTS

A follow-up meeting was scheduled for **29th January** to finalize the project topic and plan the next steps.

Next meeting

Date: 29th January 2026

Purpose: To finalize the project topic and continue work on selected datasets.

Minutes

Board meeting

DATE

February 02nd

TIME

7:00 PM

MEETING CALLED TO ORDER BY

Team Members

IN ATTENDANCE

IT Number	Name
IT24400066	Athula P.A.K. D
IT24104301	Karunagoda K.V. R
IT24101887	Fazam M.F.M
IT24101446	Wijekoon W.M.N.S. B
IT24102131	De Silva P.K. N
IT24101896	Abeywickrama J. B

APPROVAL OF MINUTES

Minutes from the previous meetings were reviewed and accepted.

BOARD

The leader guided the group to finalize the project topic and focus on AIML-based components.

REPORTS

- The project topic was finalized.
- System functions, AIML usage, non-functional requirements, and stakeholders were discussed.
- Suitable datasets were reviewed.
- Functional requirements were assigned to group members.
- Possible issues were identified, and the TADF document was completed, finalizing the first project phase.

ANNOUNCEMENTS

The group confirmed completion of the initial planning stage and agreed to proceed with implementation tasks.

Next meeting

Date: 10th February



Thank
You,

For Watching